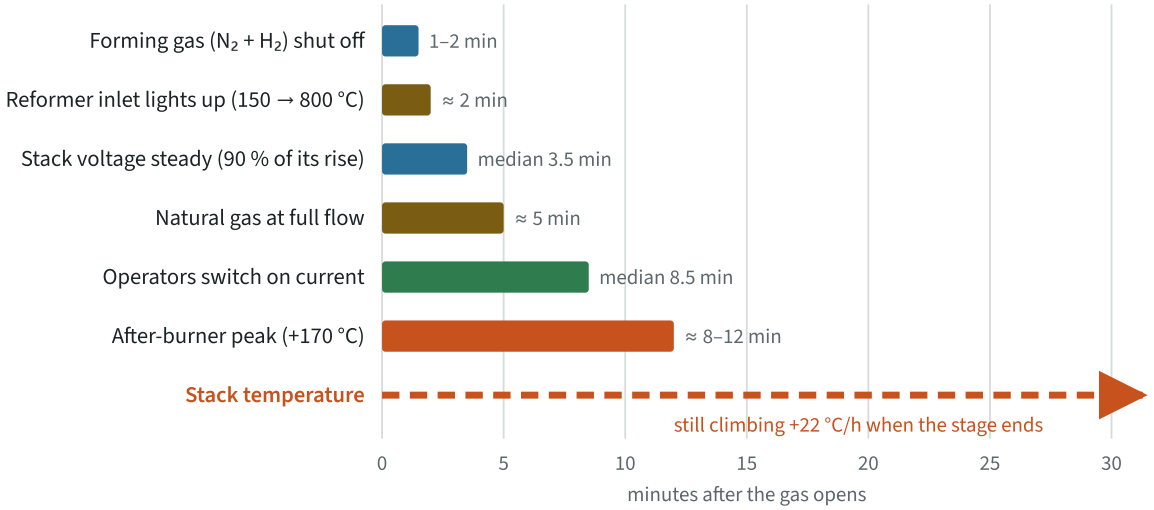
SOFC PLANT · POC DATA, MARCH 2026 · 14 FUEL SWITCHES

How the stack settles in the fuel switch

It doesn't settle, and it doesn't need to. The fuel switch is a ten-minute handover. Natural gas replaces the protective forming gas, the voltage jumps and steadies within about 4 minutes, and then the operators switch the current on. The stack temperature only changes direction: from flat to a slow climb of about +22 °C per hour. In a typical run it gains just 4 °C before the stage is over.

Why it matters: the digital twin and the data generator should treat this stage as a short, gentle ramp with a fast voltage step. They should not wait for a temperature plateau, because the real plant never waits for one.



What settles, and when. Medians of 14 times the gas was opened from hot standby (4–30 March 2026, 1-minute data). The gas path and the voltage settle within minutes; the stack temperature keeps rising until the stage ends.

PLAY THE ORDER

The numbers in one table

WHAT	BEFORE GAS (HOT STANDBY)	DURING FUEL SWITCH	RANGE OVER 14 RUNS
Stack temperature (median of 34 thermocouples)	599 °C	+4 °C gained	-2 … +14 °C
Stack warming speed	-0.4 °C/h (flat)	+22 °C/h	+9 … +47 °C/h
Stage length (gas open, no current yet)		10.5 min	3 … 29 min
Voltage per string (no load)	431 V	467 V	455 … 479 V at the top
Temperature spread inside the stack	15 °C	23 °C	13 … 56 °C at the end
Reformer inlet	148 °C	797 °C peak	740 … 824 °C
After-burner (TOX)	559 °C	728 °C peak	620 … 806 °C

Values are medians over the 14 runs. “No load” voltage means no current is drawn, so it is the open-circuit voltage. Thermocouples 19 and 25 are broken (they read 1300 °C and 54 °C) and were left out.

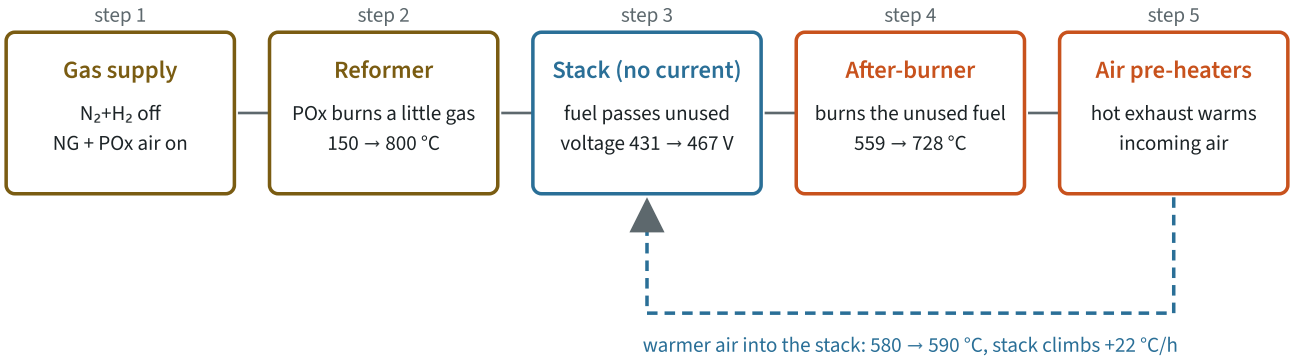
> **IF YOU REMEMBER ONE THING**
>
> The fuel switch settles the *gas and the voltage* in minutes. The stack temperature only starts to climb; it is still moving when the current comes on.

WHAT HAPPENS, MINUTE BY MINUTE →

PAGE 2 · THE REAL SEQUENCE

What happens, minute by minute

In hot standby the stack sits at about 600 °C with a protective gas around the fuel side: 1200 litres/min of nitrogen with 60 litres/min of hydrogen (“forming gas”). The fuel switch swaps that for real fuel: natural gas that a small reformer turns into a hydrogen-rich gas.

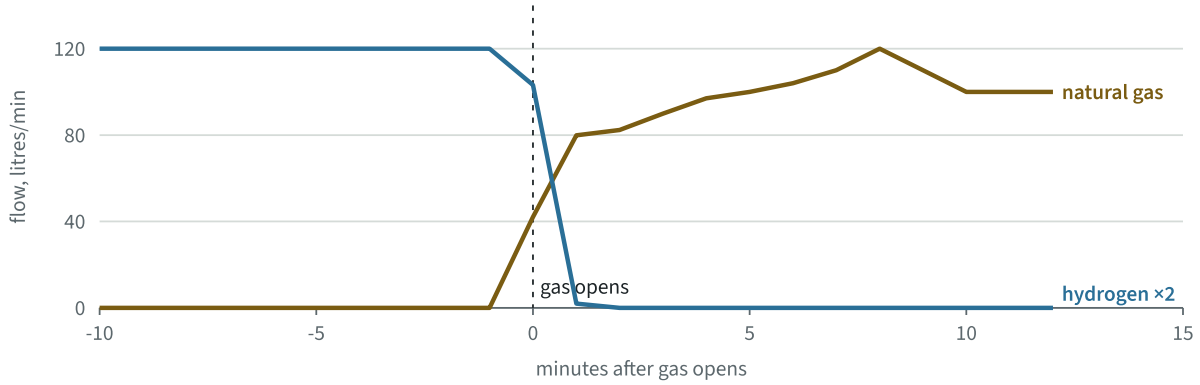


The path of the gas while no current is drawn. The cells use none of the fuel, so its energy is released in the after-burner, downstream of the stack. Only a small part of that heat comes back to the stack, through the incoming air (dashed line).

PLAY THE PATH

1. The gases swap within 2 minutes

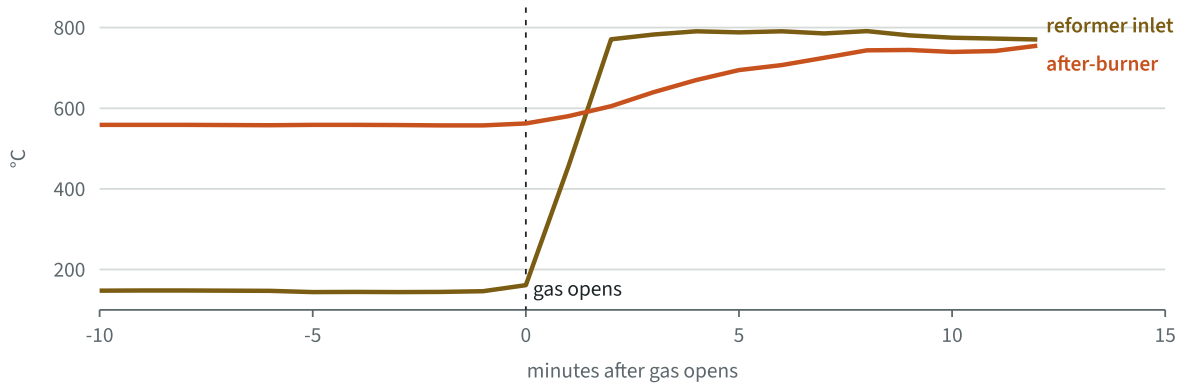
When the operator opens the natural gas (NG), the forming gas is cut at the same moment. Natural gas reaches 80–140 litres/min, usually in about 5 minutes. Partial-oxidation air (POx air, 250–620 litres/min) goes in with it; it burns a little of the gas in the reformer to supply heat.



Median flows of the 14 runs. Hydrogen is drawn twice its size so it can be seen; it falls from 60 to 0 litres/min in 1–2 minutes.

2. The reformer lights and the after-burner heats up

The reformer inlet jumps from about 150 °C to about 800 °C in 2 minutes: that is the partial oxidation starting. The fuel that reaches the stack is still about 590 °C, the same as before, so the stack gets no hot blast. With no current, the stack uses none of the fuel. It burns in the after-burner, which climbs by about 170 °C.

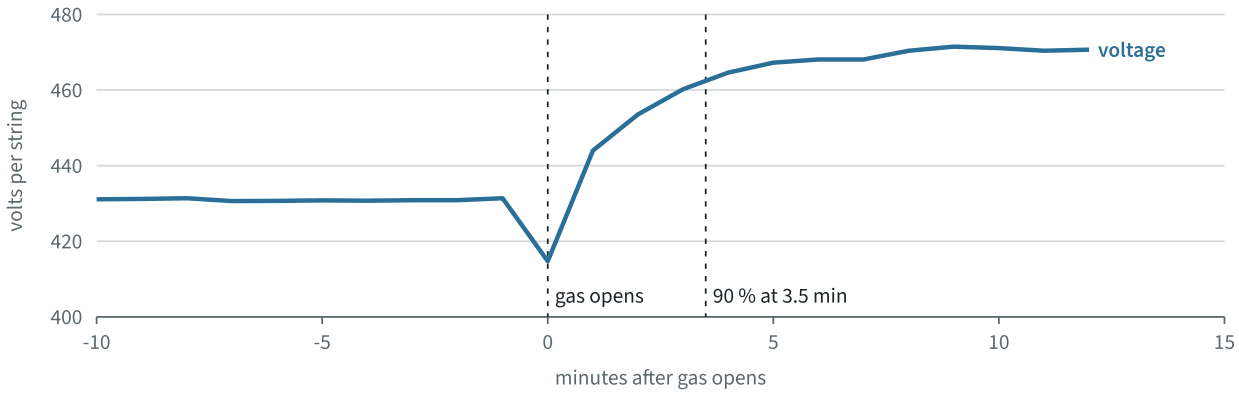


Median of the 14 runs. The after-burner peaks 8–12 minutes after the gas opens.

3. The voltage steps up and steadies

Hydrogen-rich fuel from the reformer is stronger than the thin forming gas, so the no-load voltage of each string rises from about 431 V to about 467 V. There is a short dip in the first minute, while the forming gas is gone and the reformed gas has not yet arrived. Then it reaches 90 % of the rise in a median

3.5 minutes. This steady voltage is the real “settled” signal in this stage, and it is what the operators wait for.



Median voltage of the 5 strings (TotalVoltage M1–M5), 14 runs.

IF YOU REMEMBER ONE THING

Gas swap, reformer light-off and voltage step are all done within about 5 minutes. The fuel's heat goes to the after-burner, not to the stack.

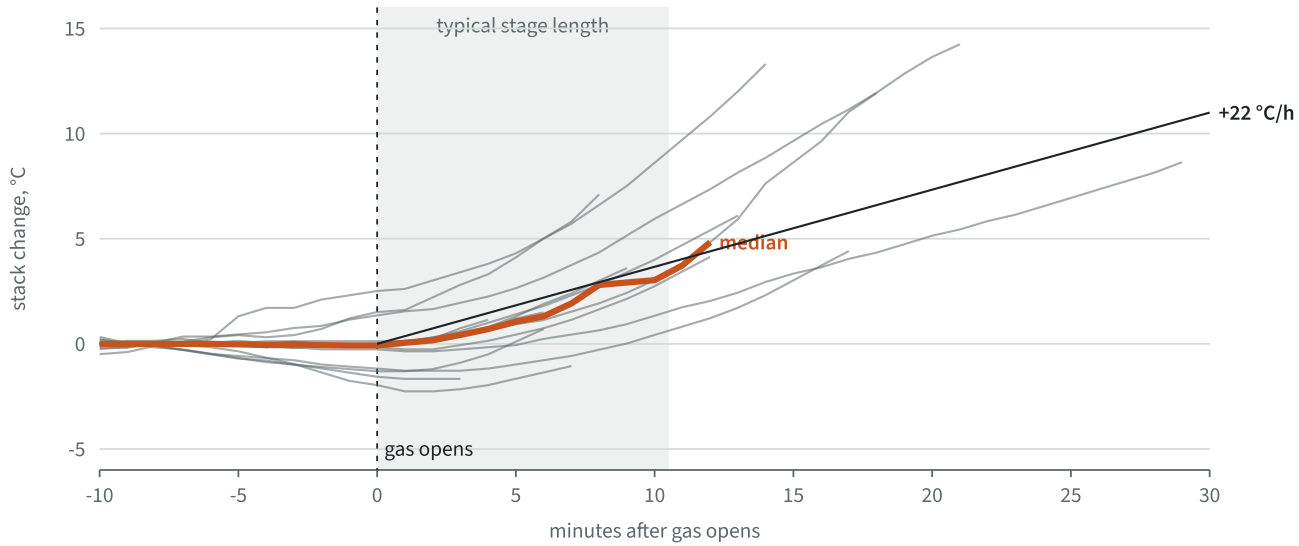
← THE SHORT ANSWER

INSIDE THE STACK →

PAGE 3 · THE STACK ITSELF

Inside the stack: a slow climb, no plateau

The stack is heavy: ceramic cells, steel plates and insulation. Think of a big cast-iron pot of water. Turn the burner up and the water starts warming at once, but it takes hours to reach a new steady temperature. The fuel switch lasts minutes.



Change of the stack temperature (median of 34 thermocouples) from its hot-standby value, each grey line one run, cut at the moment current comes on or gas closes. Before the gas the stack is flat. After it, every run bends upward; none flattens out. The orange median is drawn while at least 7 runs are still in the stage.

Speed

Flat before (−0.4 °C/h). About +22 °C/h during the stage, ranging from +9 to +47 °C/h. The speed varies a lot from run to run; the amount of gas alone does not explain it (runs at 130–140 litres/min climbed anywhere from +16 to +47 °C/h).

Gain

A median +4 °C over the whole stage. The biggest gain was +14 °C, in a 21-minute run. Even the longest run, 29 minutes on 4 March, was still rising at +20 °C/h when it ended.

Where the heat comes from

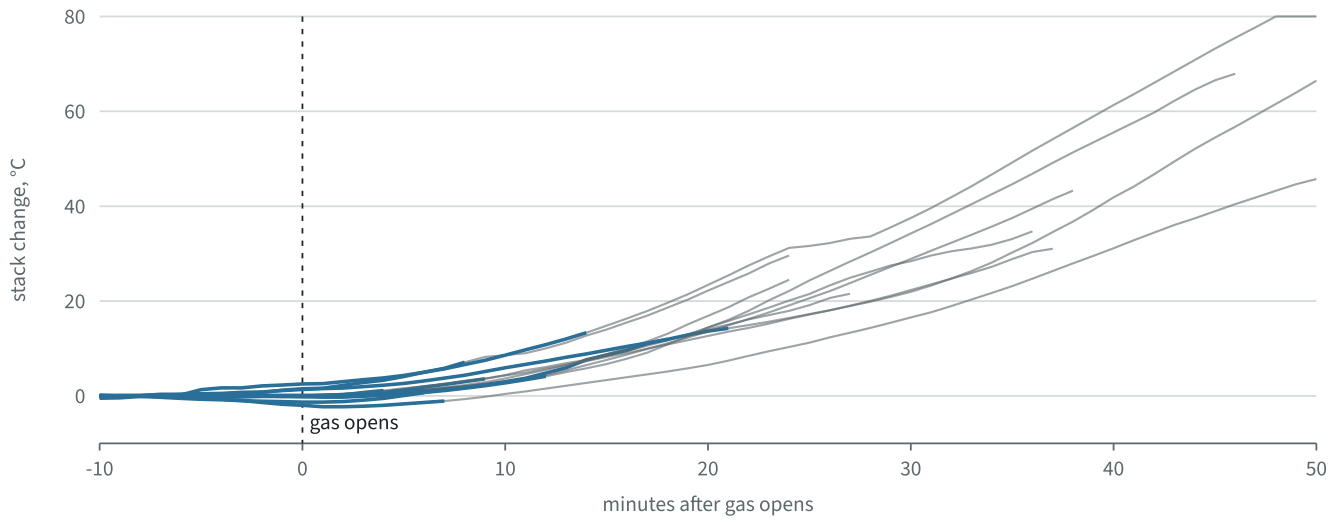
Not from the cells: with no current they make no heat. It comes from the warmer air: the after-burner's hot exhaust pre-heats the incoming air, and in 5 of 14 runs the operators also raised the air-heater target around the gas opening (for example 570 → 600 °C).

Spread inside the stack

The gap between the hottest and coldest thermocouple grows from 15 to 23 °C. Thermocouples 1–8 read almost exactly the incoming-air temperature, so they probably sit at the air inlet; the other 26 warm faster.

What happens right after

When the current comes on (part load), the cells start making their own heat and the climb speeds up about four times, to roughly +100 °C/h. That is why the operators keep part load short. The fuel switch is only the slow first bend of that curve.



The 10 runs that went on to part load, each followed until part load ended. Blue = fuel switch (gas on, no current); grey = part load (current on). The slope jumps once current flows.

IF YOU REMEMBER ONE THING

In the fuel switch the stack does not settle; it starts a slow climb of about +22 °C/h and is still climbing when the current comes on.

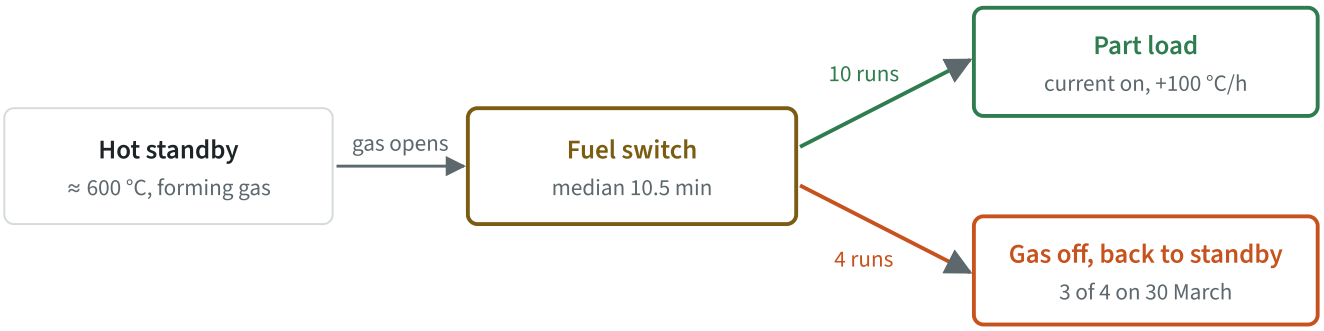
← WHAT HAPPENS, MINUTE BY MINUTE

HOW IT ENDS, ALL 14 RUNS →

PAGE 4 · HOW THE REAL PLANT ENDS THE STAGE

How it ends, all 14 runs

The plant does not end the fuel switch on a temperature. It ends it on the operator's next move: switch the current on, or close the gas.



14 gas openings from hot standby, March 2026. Two of the four “gas off” runs first tried a tiny test current (0.6 and 1.2 A) and then closed the gas. On 30 March the gas was opened four times between 08:39 and 09:53 before part load held.

GAS OPENED	MINUTES	STACK AT START °C	GAIN °C	°C/H	GAS L/MIN	VOLTS	AIR L/MIN	ENDED IN
2026-03-04 11:22	29	577	+8.6	+20	60	392→466	6000	GAS OFF → HOT STANDBY
2026-03-13 09:43	18	582	+12.0	+40	130	422→475	6000	PART LOAD
2026-03-13 13:18	8	648	+7.0	+42	140	430→467	6000	PART LOAD
2026-03-19 10:10	9	585	+3.6	+25	140	443→478	6000	PART LOAD
2026-03-19 15:24	6	605	+1.4	+16	140	450→476	6000	PART LOAD

2026-03-24 08:32	21	584	+14.2	+39	100	441→464	9999	PART LOAD
2026-03-24 13:46	7	624	-1.0	+9	80	416→456	8185	PART LOAD
2026-03-25 08:13	4	601	+1.1	+17	80	440→458	9562	PART LOAD
2026-03-25 10:11	6	614	+0.8	+20	93	431→455	9997	PART LOAD
2026-03-25 14:16	14	574	+13.2	+47	130	453→479	8894	PART LOAD
2026-03-30 08:39	13	591	+6.1	+29	80	425→456	10000	GAS OFF → HOT STANDBY
2026-03-30 09:27	3	600	-1.6	–	116	418→467	8403	GAS OFF → HOT STANDBY
2026-03-30 09:53	12	598	+4.2	+22	120	383→473	7956	PART LOAD
2026-03-30 13:01	17	620	+4.4	+20	100	436→479	6000	GAS OFF → HOT STANDBY

Minutes = gas open with no current (above 0.5 A ends the count). °C/h is a straight-line fit and is left blank when the run is shorter than 5 minutes. Volts = mean string voltage before the gas → highest during the stage.

What this means for our work

Digital twin

The twin's 10-minute wait before part load is close to the plant: a median 8.5 minutes in the 10 runs that went on to part load. It should show the voltage step of about +37 V within 3–4 minutes and a stack climb near +22 °C/h (plant range +9 to +47).

Data generator

Fuel-switch blocks should be short (3–29 min) with a rising stack, not a flat one, and an after-burner jump of 60–235 °C.

Open points for you to decide

	QUESTION	WHY IT MATTERS
A	Should “voltage steady” (90 % of its rise) be written down as the end-of-switch signal?	It is what the data shows operators waiting for; today the stage ends only when current appears.
B	Do we need to know where the stack would settle if the gas stayed on with no current?	No run lasted long enough (longest 29 min). Only a planned hold test, or the twin, can answer it.
C	Can someone confirm where thermocouples 1–8 sit?	They track the incoming air; if they are at the air inlet, the stack spread figure is an inlet-to-core gap.
D	Report thermocouples 19 and 25 as faulty?	19 reads 1300 °C (open circuit), 25 reads about 54 °C at a 600 °C stack.

> **IF YOU REMEMBER ONE THING**
>
> The real plant leaves the fuel switch when the voltage is steady, about 9 minutes after the gas opens, not when the stack temperature is steady.

← INSIDE THE STACK

BACK TO THE SHORT ANSWER